

KIDNEY STONE DISEASE WITH SPECIAL REGARD TO DRUG-INDUCED KIDNEY STONES - A CONTEMPORARY SYNOPSIS.

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Kidney stone disease (nephrolithiasis; urolithiasis) is a clinical entity with long-term course and recurrence, primarily affecting mature and ageing men, involving the formation and presence of urinary stones in the kidneys and urinary tract. The pathogenesis of this disorder is complex and still not fully understood. A rare, potentially modifiable, form of kidney stone disease takes the form of drug-induced urinary stones. The aim of the review was a brief description of the classification and pathophysiology of kidney stone disease, along with the short characteristics of drug-induced urinary stones. This type of stones is formed as a result of crystallisation in the kidneys and urinary tract of sparingly soluble drugs and their metabolites, or as a result of metabolic changes caused by drugs, predestinating the development of stones containing endogenous compounds. Conclusion: Therefore, during treatment with the use of drugs with high lithogenic potential, the safety of pharmacotherapy should be monitored in the context of its increased risk of developing urinary stones.

Urolithiasis/nephrolithiasis: what's it all about?

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Urolithiasis (urinary tract calculi or stones) and nephrolithiasis (kidney calculi or stones) are well-documented common occurrences in the general population of the United States. The etiology of this disorder is multifactorial and is strongly related to dietary lifestyle habits or practices. Proper management of calculi that occur along the urinary tract includes investigation into causative factors in an effort to prevent recurrences. Urinary calculi or stones are the most common cause of acute ureteral obstruction. Approximately 1 in 1,000 adults in the United States are hospitalized annually for treatment of urinary tract stones, resulting in medical costs of approximately dollar 2 billion per year (Ramello, Vitale, & Marangella, 2000; Tanagho & McAninch, 2004).

Better sleep quality could reinforce the relationship between sleep posture and kidney stone laterality in obstructive sleep apnea patients.

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Urolithiasis is a common urologic problem among adults worldwide. It is interesting that urinary stones tend to recur on the same side and the pathophysiology of unilateral stone formation is not clearly understood. Researchers found that sleep posture could alter renal perfusion, and subsequent vascular injury may lead to urolithiasis formation. The aim of this study is to retrospectively evaluate the correlations between the specific sleep postures recorded in polysomnography (PSG) and the stone laterality in unilateral urinary stone formers with obstructive sleep apnea (OSA). We retrospectively reviewed patients receiving PSG for the diagnosis of OSA in our institutes between January 1st 2002 and September 30th 2016. Those had a history of unilateral urinary stones which surgical intervention was warranted were included. Supine, prone, right decubitus, and left decubitus sleep postures were determined as patients remained specific postures for more than 50% of their sleep time. The laterality of stones with sleep posture was analyzed with Chi-square test. A student t test was used to assess

factors that influenced the ipsilateral stone formation. IBM SPSS Statistics 21 software was adopted to analyze the data, and p value less than 0.05 was considered statistically significant. 117 patients were enrolled in this study, among which 105 (89.7%) were male and 12 were female. The mean age at PSG examination was 53.5 ± 11.5 years. 66 patients were diagnosed as left-side urolithiasis, and 51 patients had urinary stones at right side. 58 (49.6%) patients slept predominantly in supine posture, while 53 (45.3%) patients, in decubitus posture, and no patient, in prone posture. A total of 51 patients' stones appeared at the same side as their decubitus posture ($p < 0.0001$). Shorter sleep stage I ratio ($24.3 \pm 15.5\%$ versus $52.4 \pm 24.7\%$, $p = 0.017$), longer sleep stage II + III + IV ratio ($66.2 \pm 21.9\%$ versus 30.9 ± 2.8 , $p = 0.049$), and less sleep stage change (176.6 ± 69.6 times versus 282.5 ± 91.2 times, $p = 0.041$) were correlated with stone formation on the same side. Age, gender, body mass index, and other polysomnographic parameters failed to demonstrate associations between the sides of urinary calculi and sleep postures. We discovered that sleep posture is associated with unilateral urolithiasis formation. Favorable sleep quality and less sleep stage changes could consolidate the correlations. Further prospective study is warranted to evaluate the effect of sleep posture and quality on non-OSA patients as well as the preventive effect of sleep posture modification on repeated unilateral stone formers.

The impact of Mayo Adhesion probability score on the success of extracorporeal shock wave lithotripsy for kidney stones.

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Urolithiasis is a prevalent condition in urology, with extracorporeal shock wave lithotripsy (ESWL) serving as a common treatment for kidney stones under 2 cm. The Mayo Adhesion Probability (MAP) score, calculated from perinephric fat characteristics, is traditionally used to assess surgical outcomes but has not been explored in the context of ESWL. This study aims to evaluate the effect of MAP score on the success rate of ESWL. This retrospective study included patients who underwent ESWL for kidney stones between January 2018 and June 2024 at a tertiary care center. Demographic data, stone characteristics, and MAP scores were obtained through CT imaging. ESWL success was defined as either complete stone clearance or a residual fragment less than 4 mm at three months post-procedure. Logistic regression and ROC analysis were employed to identify predictors of treatment success and to determine the optimal MAP score cutoff. A total of 260 patients were analyzed, divided into two groups: MAP score < 3 ($n = 154$) and MAP score ≥ 3 ($n = 106$). Patients with a MAP score ≥ 3 had a significantly lower ESWL success rate (49.1%) compared to those with a MAP score < 3 (71.4%; $p = 0.001$). Multivariate analysis identified MAP score, BMI, and stone size as significant predictors of ESWL success. ROC analysis established a MAP score cutoff of 2, with 76.5% sensitivity and 64.3% specificity. The MAP score is an independent predictor of ESWL success in kidney stone treatment. Preoperative MAP evaluation may improve patient selection and optimize ESWL outcomes.

Urolithiasis in the pediatric population - current opinion on epidemiology, pathophysiology, diagnostic evaluation and treatment.

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Urolithiasis, a condition in which calculi are found in the urinary tract, has been known for centuries.

Although the disease was considered casuistic in the pediatric population, its prevalence is rising among both children and infants. The occurrence of the disease is greater in developed countries, therefore urolithiasis should be considered a lifestyle disease. Its etiopathogenesis has not yet been well understood. Kidney stone formation is influenced by factors such as climate, eating habits, profession, fluid intake, genetic predisposition, urinary tract infections and malformations of the urinary tract. Calculi are usually composed of mixed mineral substances. Only about 30% are made up of one chemical substance, calcium oxalate being the most common. Urolithiasis can be asymptomatic and accidentally diagnosed. Abdominal pain is the most common clinical symptom, however disease presentation among infants is nonspecific. Hematuria is a common clinical finding. Ultrasonography is the most important diagnostic tool in the diagnosis of kidney stone disease. Metabolic evaluation is required in every case of urolithiasis in the pediatric population, as metabolic disorders can be found in the majority of cases in this age group. The spontaneous passage of calculi less than 6mm in diameter is likely. Invasive treatment should be carried out if stones exceed 6mm in diameter or fail to expulse spontaneously. Prophylactic treatment includes adequate fluid intake, healthy eating habits and physical activity to maintain a healthy weight. Urolithiasis is a recurrent disease, therefore long-term treatment, prophylaxis and a lasting change in dietary habits are essential.

Spectroscopic investigation of phase transformation of calcium oxalate dehydrates (renal calculi) using acidic *Bryophyllum pinnatum* powder.

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Urolithiasis is one of most common renal disorders, characterized by the formation of kidney stones (renal calculi) through the crystallization process within the urinary system. The frequently observed renal calculi are calcium oxalate renal calculi and treatment is done by shock wave method or lithotripsy which is harmful for other cells of the internal system. The objective of this work was to evaluate in vitro diagnosis of calcium oxalate kidney stones in the aqueous solution of *Bryophyllum pinnatum*. The *B. pinnatum* powder was mixed in apple cider vinegar and lemon juice separately to make solution 1 and 2 respectively. Apple cider vinegar and lemon juice were used as solvents due to their acidic and body compatible nature. Two surgically removed stones were dipped in solution 1 and 2. After two weeks, kidney stone of weight 2.7 g is completely dissolved in solution 2 while a considerable weight reduction of other kidney stone has been observed in solution 1. Fourier transform infrared (FTIR) spectroscopy results show the presence of two strong absorption peaks at 610 and 912 (cm⁻¹) in both solutions after dissolution of urinary stones are related to calcium oxalate dehydrate (COD). Raman spectra further confirm the dissolution of COD in solution having Raman shifts at 504 and 910 (cm⁻¹). Cluster formation and aggregation of particles has been observed in scanning electron microscopy images. This in vitro study proves that a mixture of *Bryophyllum pinnatum* powder and lemon juice is a best remedy to remove kidney stones.

Prevalence, pathophysiological mechanisms and factors affecting urolithiasis.

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The formation of urinary stone, urolithiasis, is one of the oldest known diseases affecting humans throughout different civilizations and times. The exact pathophysiological mechanism of urolithiasis is not yet clear,

as these calculi are of various types and too complex for simple understanding. A single theory cannot explain its formation; therefore, different theories are presented in various times for its explanation like free particle, fixed particle, Randall's plaque theory. In addition, various factors and components are identified that play an important role in the formation of these urinary calculi. In this review, composition of kidney stones, its prevalence/incidence, explanation of pathophysiological mechanisms and role of various factors; urinary pH, uric acid, parathyroid hormone, citrate, oxalate, calcium and macromolecules; osteopontin, matrix Gla protein, kidney injury molecules, urinary prothrombin fragment-1, Tamm-Horsfall protein, inter- α -inhibitors, have been discussed in detail.

Urolithiasis: History, epidemiology, aetiologic factors and management.

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Urolithiasis is defined as a disease diagnosed by the presence of one or more stones in the urinary tract. It is one of the oldest and most widespread diseases known to man, their discovery and characterisation chronology began with the civilisation's history. This pathology has a multifactorial aetiology, very frequent worldwide with geographic and racial variation, their prevalence is increasing in lockstep with socioeconomic development. In fact, this disorder affects between 2 and 20% of the population, with an approximate recurrence rate of 30% to 50% in 5 years. Furthermore, calciumtype stones, which are composed of calcium oxalate (CaOx) alone or a mixture of CaOx and calcium phosphate are the most common, accounting for more than 80% of cases. The medical management of urolithiasis is done by medical treatments and/or by surgical intervention for the stones extraction by the techniques such as extracorporeal shock wave lithotripsy (ESWL), ureteroscopy (URS), percutaneous nephrolithotomy (PCNL) and open surgery. However, various therapies, including thiazide diuretics and alkaline citrate, are used in an attempt to prevent stones recurrence induced by hypercalciuria and hyperoxaluria, but the scientific evidence for their effectiveness is less convincing. On the other hand, endoscopic and ESWL methods have revolutionised the treatment of urinary lithiasis, but these costly methods, can cause acute kidney injury and decreased renal function, in addition, do not prevent the probability of new stone formation. The deepening of our knowledge on all points relating to this disease is a priority for specialists in order to find adequate solutions for this disease. This review provides an overview of urolithiasis, its history, epidemiology, clinical manifestation, diagnosis and treatment methods.

Medical and surgical interventions for the treatment of urinary stones in children.

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Urolithiasis is a condition where crystalline mineral deposits (stones) form within the urinary tract. Urinary stones can be located in any part of the urinary tract. Affected children may present with abdominal pain, blood in the urine or signs of infection. Radiological evaluation is used to confirm the diagnosis, to assess the size of the stone, its location, and the degree of possible urinary obstruction. To assess the effects of different medical and surgical interventions in the treatment of urinary tract stones of the kidney or ureter in children. We searched the Cochrane Register of Controlled Trials (CENTRAL), MEDLINE (Ovid), Embase (Ovid) as well as the World Health Organization International Clinical Trials Registry Platform Search Portal and ClinicalTrials.gov. We searched reference lists of retrieved articles and conducted an electronic search for conference abstracts for the years 2012 to 2017. The date of the last search of all electronic databases was 31 December 2017 and we applied no language restrictions.

We included all randomised controlled trials (RCTs) and quasi-RCTs looking at interventions for upper urinary tract stones in children. These included shock wave lithotripsy, percutaneous nephrolithotripsy, ureterorenoscopy, open surgery and medical expulsion therapy for upper urinary tract stones in children aged 0 to 18 years. We used standard methodological procedures according to Cochrane guidance. Two review authors independently searched and assessed studies for eligibility and conducted data extraction. 'Risk of bias' assessments were completed by three review authors independently. We used Review Manager 5 for data synthesis and analysis. We used the GRADE approach to assess the quality of evidence. We included 14 studies with a total of 978 randomised participants in our review, informing eight comparisons. The studies contributing to most comparisons were at high or unclear risk of bias for most domains. Shock wave lithotripsy versus dissolution therapy for intrarenal stones: based on one study (87 participants) and consistently very low quality evidence, we are uncertain about the effects of SWL on stone-free rate (SFR), serious adverse events or complications of treatment and secondary procedures for residual fragments. Slow shock wave lithotripsy versus rapid shock wave lithotripsy for renal stones: based on one study (60 participants) and consistently very low quality evidence, we are uncertain about the effects of SWL on SFR, serious adverse events or complications of treatment and secondary procedures for residual fragments. Shock wave lithotripsy versus ureteroscopy with holmium laser or pneumatic lithotripsy for renal and distal ureteric stones: based on three studies (153 participants) and consistently very low quality evidence, we are uncertain about the effects of SWL on SFR, serious adverse events or complications of treatment and secondary procedures. Shock wave lithotripsy versus mini-percutaneous nephrolithotripsy for renal stones: based on one study (212 participants), SWL likely has a lower SFR (RR 0.88, 95% CI 0.80 to 0.97; moderate quality evidence); this corresponds to 113 fewer stone-free patients per 1000 (189 fewer to 28 fewer). SWL may reduce severe adverse events (RR 0.13, 95% CI 0.02 to 0.98; low quality evidence); this corresponds to 66 fewer serious adverse events or complications per 1000 (74 fewer to 2 fewer). Rates of secondary procedures may be higher (RR 2.50, 95% CI 1.01 to 6.20; low-quality evidence); this corresponds to 85 more secondary procedures per 1000 (1 more to 294 more). Percutaneous nephrolithotripsy versus tubeless percutaneous nephrolithotripsy for renal stones: based on one study (23 participants) and consistently very low quality evidence, we are uncertain about the effects of SWL on SFR, serious adverse events or complications of treatment and secondary procedures. Percutaneous nephrolithotripsy versus tubeless mini-percutaneous nephrolithotripsy for renal stones: based on one study (70 participants), SFR are likely similar (RR 1.03, 95% CI 0.93 to 1.14; moderate-quality evidence); this corresponds to 28 more per 1,000 (66 fewer to 132 more). We did not find any data relating to serious adverse events. Based on very low quality evidence we are uncertain about secondary procedures. Alpha-blockers versus placebo with or without analgesics for distal ureteric stones: based on six studies (335 participants), alpha-blockers may increase SFR (RR 1.34, 95% CI 1.16 to 1.54; low quality evidence); this corresponds to 199 more stone-free patients per 1000 (94 more to 317 more). Based on very low quality evidence we are uncertain about serious adverse events or complications and secondary procedures. Based on mostly very low-quality evidence for most comparisons and outcomes, we are uncertain about the effect of nearly all medical and surgical interventions to treat stone disease in children. Common reasons why we downgraded our assessments of the quality of evidence were: study limitations (risk of bias), indirectness, and imprecision. These issues make it difficult to draw clinical inferences. It is important that affected individuals, clinicians, and policy-makers are aware of these limitations of the evidence. There is a critical need for better quality trials assessing patient-important outcomes in children with stone disease to inform future guidelines on the management of this condition.

Proteomics and transcriptomics profiling reveals distinct aspects of kidney stone related

genes in calculi rats.

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Kidney stone also known as urolithiasis or nephrolithiasis, is one of the oldest diseases known to medicine, however, the gene expression changes and related kidney injury remains unclear. A calculi rat model was developed via ethylene glycol- and ammonium chloride-induction. Integrated proteomic and transcriptomic analysis was performed to characterize the distinct gene expression profiles in the kidney of calculi rat. Differential expressed genes (DEGs) were sub-clustered into distinct groups according to the consistency of transcriptome and proteome. Gene Ontology and KEGG pathway enrichment was performed to analyze the functions of each sub-group of DEGs. Immunohistochemistry was performed to validated the expression of identified proteins. Five thousand eight hundred ninety-seven genes were quantified at both transcriptome and proteome levels, and six distinct gene clusters were identified, of which 14 genes were consistently dysregulated. Functional enrichment analysis showed that the calculi rat kidney was increased expression of injured & apoptotic markers and immune-molecules, and decreased expression of solute carriers & transporters and many metabolic related factors. The present proteotranscriptomic study provided a data resource and new insights for better understanding of the pathogenesis of nephrolithiasis, will hopefully facilitate the future development of new strategies for the recurrence prevention and treatment in patients with kidney stone disease.
